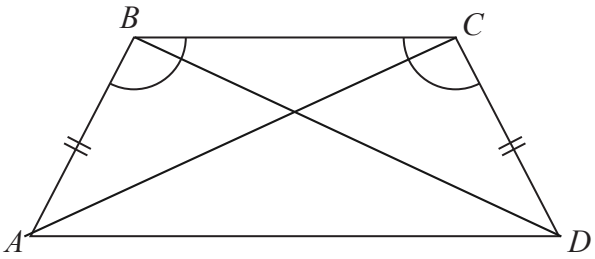
21 $ABCD$ is a quadrilateral.



$AB = CD$.
Angle ABC = angle BCD.

Prove that $AC = BD$.

Compare triangles ABC and DCB.

Angle ABC = angle DCB (given).

1 mark
Equal included angles

AB = DC (given).

1 mark
One pair of equal sides

BC = CB (the same, shared side).

1 mark
Common side, with reason

Therefore triangles ABC and DCB are congruent

by SAS (side, included angle, side).

Corresponding sides of congruent triangles are equal,

so AC = DB, as required.

1 mark
SAS and the corresponding-side conclusion

(Total for Question 21 is 4 marks)